

Case Study - Data Visualisation

Power BI - IAE Lyon 3

1. Dataset Overview

Eight datasets are provided for your use. Each one covers a distinct theme and contains **10 source files** in various formats (CSV, Excel, text, PDF). A detailed presentation sheet accompanies each dataset.

- **TechShop Global** — International e-commerce platform for technology products. Sales, customer, product, promotion and review data spanning 2 years. Enables analysis of commercial performance by country and category.
 - **Ligue 1 Football** — 2023-2024 season of the French football championship. Clubs, players, matches, individual statistics, transfers and attendance. Allows cross-analysis of sporting performance and financial data.
 - **Winter Olympics 2026** — Milano-Cortina Winter Olympic Games. Athletes, events, results, medals, ticketing and TV audiences. Enables multi-angle analysis of a major global sporting event.
 - **SoundWave** — Music streaming platform. Artists, albums, tracks, subscribers, listening data, playlists and royalties. Allows exploration of listening trends and the economics of streaming.
 - **SkyHub Airport** — International airport hub over 3 years (2022-2024). Flights, passengers, airlines, delays, cargo and satisfaction. Enables analysis of air traffic and service quality.
 - **Immobilier France** — Real estate market in major French cities over 3 years (2022-2024). Agencies, listings, transactions, energy performance diagnostics and price trends. Allows study of real estate market dynamics.
 - **Lyon Métropole** — Demographic and territorial data for the 59 municipalities over 5 years (2020-2024). Population, housing, employment, transport and taxation. Enables comparison of municipalities and their evolution.
 - **Presidential Elections** — Electoral data from the 2012, 2017 and 2022 ballots. Results, turnout, polls, campaign spending and voter demographics. Allows analysis of 10 years of French political life.
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2. Case Presentation

You will work in **groups of 2 students**. Your mission is to choose one dataset from the 8 available, explore its data, clean and model it in Power BI, then build a **visual report of approximately 5 pages** that tells a coherent story from the data.

Data quality is **intentionally imperfect**: inconsistent formats, missing values, different encodings, unstructured data. It is your responsibility to identify and address these issues.

3. General Organisation

Dataset Selection Rules

- **First come, first served:** the first group to choose a dataset has priority
- A dataset can only be assigned to **one group**

Required Tool

Power BI Desktop is the sole tool for this project. All steps (import, cleaning, modelling, visualisation) must be carried out in Power BI.

4. Learning Objectives

- Import and use the **majority of the data sources** provided
- Clean and transform data using **Power Query**
- Build a **coherent and structured data model**
- Create **DAX measures** and relevant indicators
- Design a Power BI report applying the **fundamental principles of data visualisation**
- Build a **clear storytelling narrative**: do not show everything, but tell a story
- Use **different types of visuals** appropriate to the data presented

5. Project Workflow

1. **Data Exploration** — Open each file, understand its content, identify columns, formats and potential quality issues.
2. **Cleaning and Transformation** — Using Power Query: correct data types, handle encodings, deal with missing values, remove duplicates, standardise formats.
3. **Modelling** — Structure the tables, define relationships between them, organise the model logically.
4. **Indicator Creation** — Define the DAX measures needed for your analysis. Focus on indicators that support your storytelling.
5. **Report Building** — Design approximately 5 pages with a logical progression. Each page should have a clear purpose. Pay attention to formatting and readability.

6. Expected Deliverables

A. Power BI Report

A **.pbix** file containing approximately **5 pages** of visualisations. The report must be self-contained: a reader should be able to understand it without any oral explanation.

B. Oral Presentation

Presentation during the final class session. Duration: **10 to 15 minutes**. Both group members must speak. You will present your approach, your choices and your results using your Power BI report as support.

C. Methodology Report

A document of approximately **10 pages** explaining:

- The **overall approach** adopted
- The **transformations** performed in Power Query and why
- The **data model** chosen (diagram + justification)
- The **visual choices** and the logic behind each page
- The **storytelling**: what narrative thread, what progression
- The **limitations** of your analysis and the difficulties encountered

7. Power BI and Data Visualisation Expectations

Visual Readability

- Clear titles on every page and every visual
- No clutter: each visual must provide useful information
- Legible font sizes, adequate spacing
- Measured use of colours

Graphic Consistency

- Consistent colour palette throughout the report
- Uniform visual style (fonts, sizes, alignment)
- Smooth navigation between pages

Storytelling

- Each page should answer a question or illustrate an analytical angle
- Logical progression of the report: from general to specific
- Do not try to show everything: **select the most relevant analyses**

Common Mistakes to Avoid

- Stacking visuals without a narrative thread
- Using a chart type unsuitable for the data
- Overloading a page with too much information
- Neglecting data cleaning (types, formats, outliers)
- Leaving unnecessary columns or tables in the model
- Forgetting filters, slicers or interactions between visuals
- Using inconsistent or too many colours
- Not testing the report's readability on a different screen

8. Evaluation Criteria

Criterion	Details
Import and Cleaning	Sources used, quality of Power Query transformations
Modelling	Model consistency, relevant relationships, DAX measures
Report Quality	Readability, graphic consistency, variety of visuals
Storytelling	Clear narrative thread, logical progression, relevance of analyses
Oral Presentation	Clarity, ability to explain choices, balanced participation
Methodology Report	Completeness, writing quality, critical thinking (limitations)